Yifei Zhang

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Education

Emory University Aug. 2022 – Mar. 2026
- Ph.D. in Computer Science
- Advisor: Prof. Liang Zhao
- Dissertation: Benchmarking, Correcting, and Accelerating AI Reasoning across Diverse Modalities

Columbia University Sept. 2020 – Dec. 2021
- M.S. in Data Science

Duke University Aug. 2019 – May 2020
- Master of Engineering Management

Dalian University of Technology Sept. 2015 – June 2019
- B.E. in Engineering Mechanics

Research Highlights

- **AI Agents and Evaluation.** Develop benchmarks and evaluation methods for long-horizon dialogue, goal tracking, memory, and multimodal assistant behavior, including ATOD and Hear2Act.
- **Explainable and Trustworthy AI.** Build explanation-guided learning methods and benchmarks for evaluating visual explanations, including MAGI, ELAD, and Saliency-Bench.
- **Multimodal Reasoning.** Study retrieval-augmented image generation, cognitive-emotional reasoning, and agentic medical reasoning across language, vision, and clinical data.

Research and Industry Experience

Amazon Apr. 2026 – Present
Applied Scientist Boston, MA
- Develop scalable evaluation methods for agentic conversational systems, covering reliability, task completion, conversational quality, and multimodal behavior.
- Combine diverse evaluation signals into interpretable quality assessments for model and system improvement.
- Research privacy-aware model adaptation and distillation under practical system constraints.

Amazon May 2025 – Aug. 2025
Applied Scientist Intern Boston, MA
- Developed a memory system and multi-agent evaluation framework for agentic conversational systems.
- Built synthetic dialogue-generation and benchmarking pipelines for long-horizon, multi-turn, and concurrent user tasks.

Amazon AGI June 2024 – Sept. 2024
Research Scientist Intern Boston, MA
- Developed reward modeling and preference-based evaluation workflows for LLM-driven conversational interactions.
- Built synthetic preference-data generation and parameter-efficient fine-tuning pipelines.

Emory University Aug. 2022 – Mar. 2026
Ph.D. Student Researcher Atlanta, GA
- Developed evaluation benchmarks and explanation-guided methods for reliable AI across language, vision, and multimodal data.

Publications

Full publication list: Google Scholar.

Peer-Reviewed and Accepted Papers

Dual Decomposition for Text-to-Image Retrieval and Generation EACL Findings, 2027 (accepted)

- M. Zhu, S. Cheng, G. Bai, **Yifei Zhang**, and L. Zhao

Beyond Efficiency: A Systematic Survey of Resource-Efficient Large Language Models ACM CSUR, 2026

- G. Bai, Z. Chai, C. Ling, S. Wang, J. Lu, N. Zhang, T. Shi, Z. Yu, M. Zhu, **Yifei Zhang**, X. Song, C. Yang, Y. Cheng, and L. Zhao

ATOD: An Evaluation Framework and Benchmark for Agentic Task-Oriented Dialogue Systems AACL-IJCNLP, 2026

- **Yifei Zhang**, H. Nayyeri, R. Khaziev, E. Yilmaz, G. Tur, D. Hakkani-Tür, and H. P. Thadakamalla

Hear2Act: Benchmarking When Prosody Should Change What an Assistant Does NeurIPS RTCA, 2026 (workshop poster)

- X. Liu, H. Nayyeri, D. Hakkani-Tür, E. Yilmaz, J.-K. Kim, **Yifei Zhang**, C. Peris, and H. Thadakamalla

Saliency-Bench: A Comprehensive Benchmark for Evaluating Visual Explanations KDD, 2025

- **Yifei Zhang**, J. Song, S. Gu, T. Jiang, B. Pan, G. Bai, and L. Zhao

GraphNarrator: Generating Textual Explanations for Graph Neural Networks ACL, 2025

- B. Pan, Z. Xiong, G. Wu, Z. Zhang, Y. Hu, **Yifei Zhang**, and L. Zhao

Distilling Large Language Models for Text-Attributed Graph Learning CIKM, 2024

- B. Pan, Z. Zhang, **Yifei Zhang**, Y. Hu, and L. Zhao

Multi-Prompt Fine-Tuning of Foundation Models for Enhanced Biomedical Image Segmentation Springer, 2024 (book chapter)

- X. Li, **Yifei Zhang**, and L. Zhao

DUE: Dynamic Uncertainty-Aware Explanation Supervision via 3D Imputation KDD, 2024

- Q. Zhao, **Yifei Zhang**, M. Zhu, S. Gu, Y. Gao, X. Yang, and L. Zhao

ELAD: Explanation-Guided Large Language Models Active Distillation ACL Findings, 2024

- **Yifei Zhang**, B. Pan, C. Ling, Y. Hu, and L. Zhao

Visual Attention Prompted Prediction and Learning IJCAI, 2024

- **Yifei Zhang**, B. Pan, S. Gu, G. Bai, M. Qiu, X. Yang, and L. Zhao

STES: A Spatiotemporal Explanation Supervision Framework SDM, 2024

- D. Yu, B. Chen, Y. Li, S. Dhakal, **Yifei Zhang**, Z. Liu, M. Zhang, J. Zhang, and L. Zhao

MAGI: Multi-Annotated Explanation-Guided Learning ICCV, 2023

- **Yifei Zhang**, S. Gu, Y. Gao, B. Pan, X. Yang, and L. Zhao

ESSA: Explanation Iterative Supervision via Saliency-Guided Data Augmentation KDD, 2023

- S. Gu, **Yifei Zhang**, Y. Gao, X. Yang, and L. Zhao

Preprints

Agentic Commerce: A Survey of How AI Agents Are Reshaping Commerce TechRxiv, 2026
- **Yifei Zhang**, B. Pan, M. Zhu, J. Pei, and L. Zhao

EmoLLM: Appraisal-Grounded Cognitive-Emotional Co-Reasoning in Large Language Models arXiv, 2026
- **Yifei Zhang**, M. Li, H. Gao, and L. Zhao

Explainable Cross-Disease Reasoning for Cardiovascular Risk Assessment from Low-Dose Computed Tomography arXiv, 2025
- **Yifei Zhang**, J. Zhang, M. Safari, X. Yang, and L. Zhao

MEGL: Multimodal Explanation-Guided Learning arXiv, 2024
- **Yifei Zhang**, T. Jiang, B. Pan, J. Wang, G. Bai, and L. Zhao

Saliency-Guided Hidden Associative Replay for Continual Learning arXiv, 2023
- G. Bai, Q. Zhao, X. Jiang, **Yifei Zhang**, and L. Zhao

SurroCBM: Concept Bottleneck Surrogate Models for Generative Post-hoc Explanation arXiv, 2023
- B. Pan, Z. Liu, **Yifei Zhang**, and L. Zhao

Controllable Data Generation Via Iterative Data-Property Mutual Mappings arXiv, 2023
- B. Pan, M. Qin, S. Wang, **Yifei Zhang**, and L. Zhao

Academic Service

Journal Reviewer: ACM TIST; IEEE TBD; IEEE TNNLS; JACMP; Expert Systems with Applications; Information Processing and Management; Neurocomputing; Information Systems Journal.

Conference Reviewer / Program Committee: ICLR 2025–2027; NeurIPS 2026; AAAI 2027; CVPR 2025–2026; ICCV 2025; ECCV 2026; BMVC 2026; ACL ARR 2025–2026; KDD Research Track 2025–2026; KDD Datasets and Benchmarks Track 2025–2026; MemAgents Workshop (ICLR 2026).

Workshop Organization: Publicity Chair, DeepSpatial 2024 (KDD); Student Co-organizer, FM4sBio 2025 (AAAI).

Honors, Grants, and Memberships

- Sigma Xi, Full Membership.
- Tinker Research Grant.
- KDD 2023 Student Travel Award.
- Dalian University of Technology Overseas Alumni Association Seagle Scholarship Award.
- Academic Excellence Scholarship Award, Dalian University of Technology.
- Technological Innovation Scholarship Award, Dalian University of Technology.

Technical Skills

Programming: Python, C/C++, MATLAB, Java, Shell
AI / LLM Frameworks: PyTorch, Transformers, TensorFlow, vLLM, verl, TRL, PEFT
Training and Infrastructure: Accelerate, DeepSpeed, AWS, Docker, Linux, Git